

Jayden Melendez

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EDUCATION

Marist College, Poughkeepsie, New York
B.S. in Computer Science, Minor: Cyber Security

Graduation: May 2026
GPA: 3.95

RELEVANT COURSEWORK

Artificial Intelligence in Software Development, Operating Systems, Algorithms, Computer Organization and Architecture, Intro to z/OS & Major Subsystems, Internetworking, Database Management, Mobile Security

PROFESSIONAL EXPERIENCE

International Business Machine (IBM), Poughkeepsie, New York
Backend Developer Co-op

May 2024 – Present

- **Architectural Modernization:** Contributed to the refactoring of a legacy **50,000+ LoC Java** mainframe patch management system; transitioned tightly-coupled modules into a **Service-Oriented Architecture (SOA)** to improve horizontal scalability and maintainability.
- **Systems Engineering:** Engineered a programmatic diagnostic engine that parses data across hundreds of relational database tables to automate hardware issue detection; reduced manual troubleshooting time by **10%** for 50+ global support.
- **Performance Optimization:** Optimized complex **SQL queries and indexing strategies** within a DB2 environment, achieving a **20% reduction** in existing data retrieval latency for critical patch lifecycle tracking.
- **Reliability & DevSecOps:** Enhanced system stability by implementing a comprehensive **SwaggerUI** testing suite and utilized **Podman** for containerized development, ensuring codebase environmental consistency among team members.
- **Test-Driven Development:** Adopted **TDD** using **JUnit** to ensure the reliability of the refactored SOA modules, reducing production regressions during the mainframe modernization.
- **Collaborative Leadership:** Actively collaborate with senior architects to identify technical debt, utilizing **Agile methodologies** and **Github Enterprise** for code reviews and version control to ensure high-quality software delivery.

Marist College, Poughkeepsie, New York
Resident Assistant

Aug 2023 – Present

- Manage crisis intervention, conflict resolution, and facility logistics for a community of 60+ residents.
- Facilitate community engagement and budget management through coordinated educational programming.

PROJECTS

F.A.S.T — Financial Account Statement Transformer

Feb 2026 – Present

Python, Scikit-learn, Docker, Jenkins, Pydantic, Ollama Phi-4 Mini, YAML

- Engineering a local, containerized Python DAG pipeline to automate the extraction and standardization of bank statement data from PDFs into unified CSV schemas using Ollama's local LLM to parse the data from the PDFs.
- Implementing a machine learning router using Scikit-learn (TF-IDF and SGDClassifier) to perform context-aware transaction classification and deduplication across different account types.
- Architecting a robust data validation layer with Pydantic, featuring two-tier reconciliation (mathematical balance checks and sign validation) to ensure 100% data integrity.
- Developing a modular, configuration-driven system using Docker and YAML, enabling seamless onboarding of new bank formats without modifying core logic.
- **CI/CD & Quality Assurance:** Implementing a CI/CD pipeline using **Jenkins** to automate unit and module testing suites, ensuring **100% code coverage** for the Scikit-learn router and Pydantic validation layers before containerized deployment.

NutritionU — A meal planner for students

Aug 2025 – Dec 2025

React, Vite, Node.js, Postgres, Postman

- **Full-Stack Development:** Built a full-stack meal planning application using **React**, **Node.js**, and **PostgreSQL**, enabling students to generate personalized, budget-conscious meal plans based on dietary goals.
- **AI Integration:** Integrated the **OpenAI API** to power intelligent recipe substitution for dietary restrictions, automated macro-nutrient target calculation, and dynamic meal plan generation from user grocery lists.
- **Database Design:** Architected a normalized **PostgreSQL** schema to manage user profiles, grocery inventories, and nutritional data, enforcing data integrity through foreign key constraints and parameterized queries to prevent SQL injection.

SKILLS

Programming Languages: JavaScript, SQL, Python, Java, TypeScript, C++, Assembly, HTML/CSS

Technologies/Frameworks: React, ReactNative, Springboot, Python FastAPI, Node.js, Docker, Linux

Developer Tools: VS Code, GitHub, SwaggerUI, Cursor, Claude Code

EXTRACURRICULARS

Ultimate Frisbee Player, Marist College, Poughkeepsie, NY

Aug 2022 – Present

Spiritual Life and Service Volunteer, Marist College, Poughkeepsie, NY

Aug 2022 – Present